



Paper Id: 250755

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Subject Code: RCAI402

Roll No:

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MCA-INT
(SEM IV) THEORY EXAMINATION 2024-25
DATA STRUCTURES USING C

TIME: 3 HRS**M.MARKS: 70**

Note: 1. Attempt all Sections. If require any missing data, then choose suitably.

SECTION A

1. Attempt all questions in brief.

2 x 7 = 14

a.	Differentiate linear and non-linear data structure.
b.	Define ADT (Abstract Data Type).
c.	What are the advantages of linked list?
d.	What is circular queue?
e.	What is priority queue? Give the applications of priority queue.
f.	What is a threaded binary tree? Write the advantages of threaded binary tree.
g.	Differentiate between directed graph and undirected graph?

SECTION B

2. Attempt any three of the following:

7 x 3 = 21

a.	Write a C program to find the length of a string without using strlen().
b.	Write an algorithm to evaluate postfix expression using stack. And also evaluate following postfix expression by using stack: 6 2 3 + - 3 8 2 / + * 2 ↑ 3 +
c.	For a binary search tree T, its pre-order and in-order traversals are as follows: Pre-order : A B L M K N P Q In-order : L B M A N K Q P Draw the binary search tree T.
d.	What is binary search tree (BST)? Write an algorithm for inserting a given value in BST.
e.	Define spanning tree of a graph. What is minimum spanning tree? Discuss any one method of finding the minimum spanning tree of a graph.

SECTION C

3. Attempt any one part of the following:

7 x 1 = 7

(a)	Write an algorithm for converting infix expression to postfix expression using stack.
(b)	What do you know by asymptotic notations? Define different kinds of asymptotic notations.

4. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain the insertion operation in linked list. How nodes are inserted after a specified node.
(b)	What is doubly linked list? Write an algorithm to perform insertion on a doubly linked list.



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5. Attempt any *one* part of the following:

7 x 1 = 7

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|-----|---|
| (a) | Draw the binary tree for following arithmetic expressions:
1) $E = (a / c) + (b * d)$
2) $E = (((a * b) * c) * d)$ |
| (b) | Discuss the array representation of binary tree with an example. Also briefly discuss various traversal methods of binary tree. |

6. Attempt any *one* part of the following:

7 x 1 = 7

- | | |
|-----|---|
| (a) | Write down Quick Sort algorithm and apply it on following data to sort it:
5, 8, 7, 6, 3, 4, 1, 9, 2, 10 |
| (b) | What is AVL tree? Discuss the balancing factor? |

7. Attempt any *one* part of the following:

7 x 1 = 7

- | | |
|-----|---|
| (a) | Briefly explain basic operations that can be performed on a file. Explain indexed and sequential file organization. |
| (b) | Write down algorithm for Depth -First search in the graph. Apply that algorithm on given graph to compute DFS Tree. |

